Models of Interception Movements

From Raw Trajectories to Individual Behavioural Signatures

Preliminary Results

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1. Overview and goal

The goal of this project is to compress each interception movement into a small set of numbers — a low-dimensional latent code — that can (a) reconstruct the movement and (b) act as an individual "fingerprint" whose aggregate over a subject's trials predicts their behaviour. We build up through three phases: a K-Means clustering check (is any subject structure present at all?), a polynomial-spline baseline (a non-machine-learning reference), and a Conditional Variational Autoencoder (CVAE), the core model. All numbers below are evaluated on **7 subjects held out entirely from training**, so every score measures generalisation to new people. All results are averaged over 10 random train/test splits (seeds), to control for the small number of subjects.

2. Data and trial segmentation

We analyse the free eye-movement condition (condition 2): 3D finger trajectories at 240 Hz. We segment each trial by **task events** rather than by a velocity threshold. The target appears, holds still for a randomised foreperiod (0.18–0.48 s), then **starts moving** (the go-signal); the participant may only move after that. We take the **go-signal as the behavioural zero-time** (so reaction time is measured from it and the randomised foreperiod does not leak into the data), and the window **ends at finger arrival** (interception). Each movement is low-pass filtered (10 Hz), resampled to 100 frames, and origin-aligned.

From 4,763 condition-2 trials we retain **4,684 (98.3%)**, dropping 48 "too early" (moved before the go-signal), 27 that arrive >1 s after the target's window (disengagement), and 4 timeouts. The event-based window also removed a segmentation artefact in which late sensor jitter had stretched a 0.5 s reach to as long as 9.7 s; the maximum movement time is now 1.36 s.

3. Phase 1 — K-Means baseline: is the signal even there?

K-Means is a classical clustering algorithm: it groups trials by geometric similarity, with no notion of "subject." We check whether the clusters line up with the true subjects (Adjusted Rand Index, ARI: 1.0 = perfect match, 0 = random).

Representation	ARI	NMI
Raw trajectories (300-dim)	0.093	0.292
Kinematic features (11-dim)	0.104	0.294

Interpretation. Both scores are weak but clearly above chance (0). A little subject structure exists, but a simple algorithm barely recovers it. This is a *good* baseline result: it confirms individual signatures are real yet hard to extract, which is exactly what motivates a deep generative model. (A high K-Means score would undermine the need for one.)

4. Phase 2 — Polynomial spline baseline

The spline is pure mathematics (no learning): we fit a smooth cubic curve to each trajectory. It copies the *shape* almost perfectly — a per-trial fit reaches a reconstruction MSE of ~0.0002, an interpolation ceiling — and even a capacity-matched version (Spline+PCA, reduced to the same number of dimensions) reconstructs geometry slightly better than the CVAE (0.20 vs 0.28 at n=3). **But a spline's coefficients describe a line, not a person.** On the quantities we care about (timing, curvature for unseen subjects, and generating realistic movement distributions), the CVAE beats it significantly. The spline is a better *tracer*; the CVAE is a better *portrait painter*. This is why reconstruction MSE alone is the wrong scoreboard for a fingerprinting project.

5. Phase 3 — the Conditional VAE (our model)

What it is and what it outputs

The CVAE has an encoder that squeezes a trajectory into n latent numbers, and a decoder that rebuilds it. We chose a **variational** autoencoder (not a plain one) because its latent space is smooth and continuous — which is what makes averaging a subject's trials into one meaningful fingerprint valid. It is **conditional**: we feed the task facts (start position, side) separately, so the latent is free to encode personal style rather than the task. The decoder has two outputs: **the full reconstructed trajectory** (its main job) **and a timing head** that predicts movement time and reaction time — the temporal axis that resampling removes (following Prof. Friedman's suggestion). So a generated code yields a shape *and* its durations.

Latent-dimension sweep

We swept the fingerprint size n over {2, 3, 4, 8, 16} (3 seeds each).

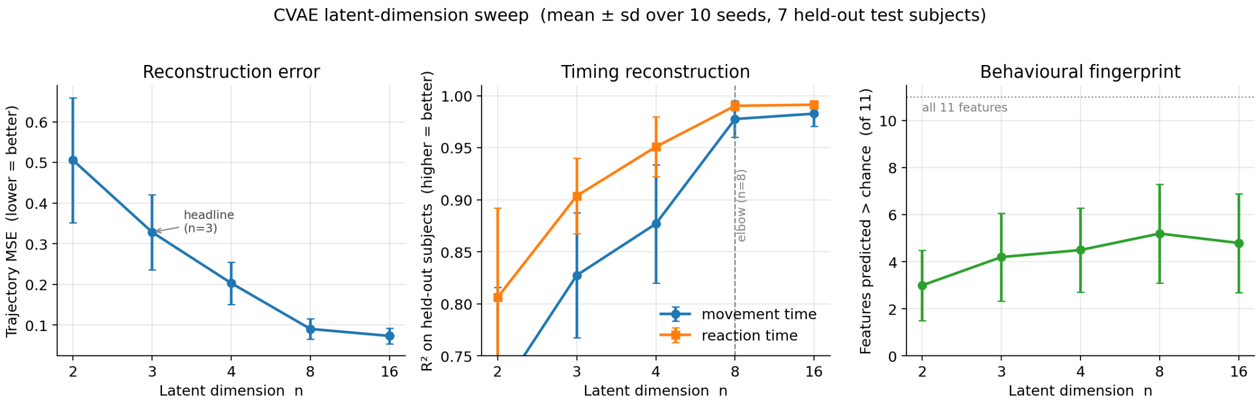

Figure 1. As n grows, reconstruction improves, timing prediction saturates around the n=8 elbow (R^2 ~ 0.98), and the behavioural fingerprint peaks around n=8.

n	Recon MSE	Timing R^2 (move)	Timing R^2 (react)	Behav. features > chance (of 11)
2	0.51	0.71	0.81	3.0
3	0.33	0.83	0.90	4.2
4	0.20	0.88	0.95	4.5
8	0.09	0.98	0.99	5.2
16	0.07	0.98	0.99	4.8

What we chose and why. We report all n. **n = 8 is the sweet spot**: reconstruction and timing saturate there (R^2 ~ 0.98) and the behavioural fingerprint also peaks (~5 of 11 features), with no gain at n = 16. We nonetheless take **n = 3 as the headline fingerprint** for presentation — the smallest size that is directly visualizable, already capturing timing at R^2 ~ 0.83–0.90. **Compared with the spline**, the CVAE gives up a little point-wise geometry in exchange for a structured, interpretable latent space and far stronger timing and behavioural prediction — the properties the fingerprint needs.

What each latent variable controls

Because n=3 is small, we can ask what each latent number does by correlating it with movement features (Figure 2). All three axes turn out to encode **timing and speed** (reaction time, movement time, peak speed) in different combinations; none cleanly isolates spatial shape such as curvature. In other words, the fingerprint the model learns is dominated by *temporal* dynamics. This is useful and honest: it tells us the latent is interpretable but timing-centric, and points to spatial/strategy features (e.g. number of sub-movements) as something a future version could encode more explicitly.

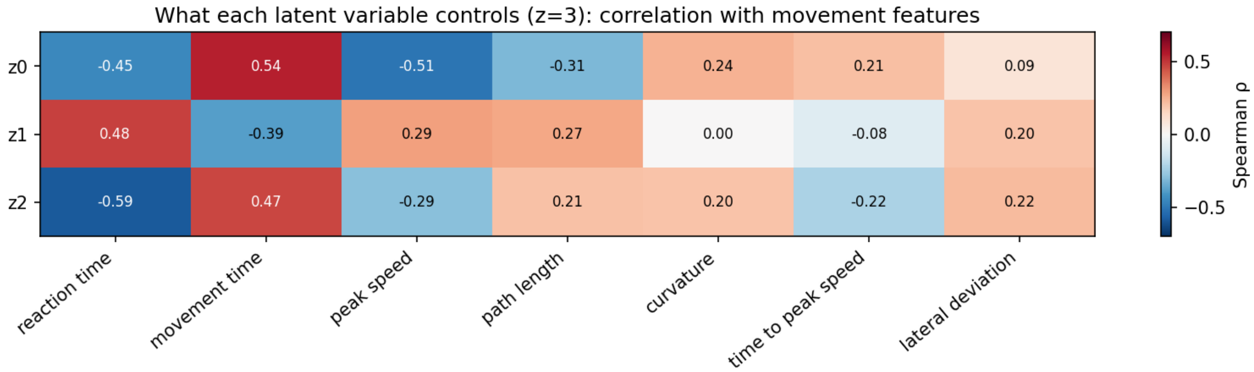

Figure 2. Correlation of each latent (z0–z2) with movement features. Red = positive, blue = negative. The axes are timing/speed-dominated.

6. Subject fingerprints

A subject **fingerprint** is the aggregate of that subject's trial latent codes — the mean position (who they are on average) plus the spread (their trial-to-trial variability). Probing from the fingerprint to a subject's behaviour, for unseen subjects, the number of behavioural features predicted above chance peaks around 5 of 11 at n=8 (with no improvement at n=16), but per-feature accuracy stays weak and subjects do not separate cleanly.

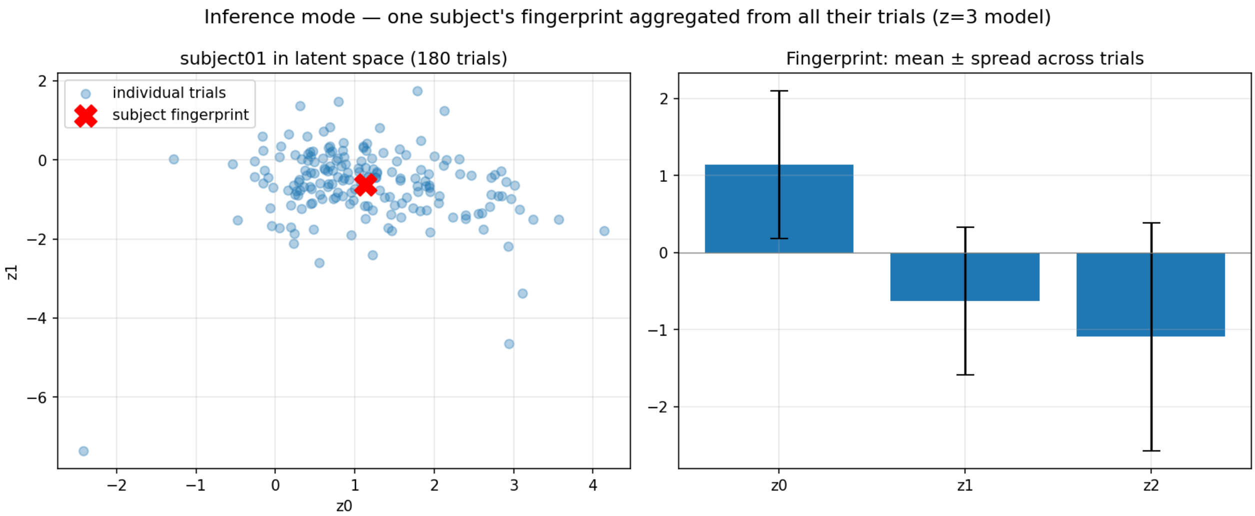

Figure 3. One subject's 180 trials (blue) and their fingerprint (red X). The wide cloud — within-subject variability larger than between-subject differences — is precisely why clean separation is hard.

This weakness is a data limitation, not a modelling failure. It persists across every fingerprint size, and an independent comparison found a 1,700×-larger model and a small PCA model plateauing at the same separation. With only 28 subjects, and trial-to-trial variability larger than the differences between people, the individual signature is only weakly recoverable. We therefore present it as a validated finding and lead with the strong kinematic and timing results.

7. The dashboard (deliverable)

We built an interactive tool on the trained model with two modes. **Exploration** lets a researcher move the latent sliders and watch a movement (and its timing) be generated — a way to see what the fingerprint space contains. **Inference** takes a subject's trials and reads out their fingerprint (mean + spread), reproducing Figure 3 for any uploaded subject.

Figure 4. Exploration mode: four latent settings produce four different generated movements, each with its own movement and wait time.

8. Where we followed the proposal, and where we deviated

For transparency, this compares what the submitted proposal said we would do against what we did.

Proposal item	Status	Note
K-Means baseline (2 representations)	Done	Raw trajectory + kinematic features.
Spline baseline (cubic, few knots)	Done	As specified.
CVAE + timing head	Done	Predicts movement time and reaction time.
Latent sweep n = 2,3,4,8,16	Done	Full sweep; headline n=3, elbow n=8.
Eval: MSE, Spearman, probing R^2, KS, MMD, energy	Done	All implemented.
Dashboard (Inference + Exploration)	Done +	Extended to multi-trial subject fingerprints.
Segmentation of the movement window	Refined	Changed to event-based (go-signal to arrival), beyond the proposal, after invest
Trial filtering	Added	Drop too-early / >1 s-late / timeout using the .mat outcome labels (not described
Benchmark vs Friedman sub-movement pipeline	Not done	External dependency not yet available; pending (see questions).
Fallback to a supervised predictor	Not needed	The VAE converged; the fallback was not triggered.

9. Methods summary

- **Data:** condition 2, 4,684 trials, 28 subjects; 3D finger position at 240 Hz.
- **Preprocessing:** event-based segmentation (go-signal to arrival), 10 Hz Butterworth low-pass, resample to 100 frames (cubic spline), origin normalisation.
- **Split:** leave-N-subjects-out, 17 train / 4 validation / 7 test.
- **Model:** Conditional VAE, encoder/decoder MLP (hidden 256), condition = start position + side, timing head predicting movement & reaction time; KL annealing; standard-ELBO loss.
- **Baselines:** K-Means (trajectory & feature), polynomial spline (per-trial & Spline+PCA).
- **Evaluation (7 held-out subjects):** reconstruction MSE; timing R^2; latent–feature Spearman correlations; behavioural probing (linear + SVR, leave-one-subject-out); generative fidelity (KS, MMD, energy distance).
- **Tools:** Python, PyTorch, scikit-learn, NumPy/Pandas, Matplotlib, Streamlit.

10. Architecture & feature comparison (results)

We implemented and tested the three natural extensions against the baseline, across all latent sizes and 10 seeds (Figure 5): a **1-D convolutional** encoder/decoder (which reads the trajectory as a time-ordered signal rather than an unordered vector), conditioning on the **exact target speed** (rather than only its range), and adding the **sub-movement count** (peaks in the speed profile) as a feature and probing target.

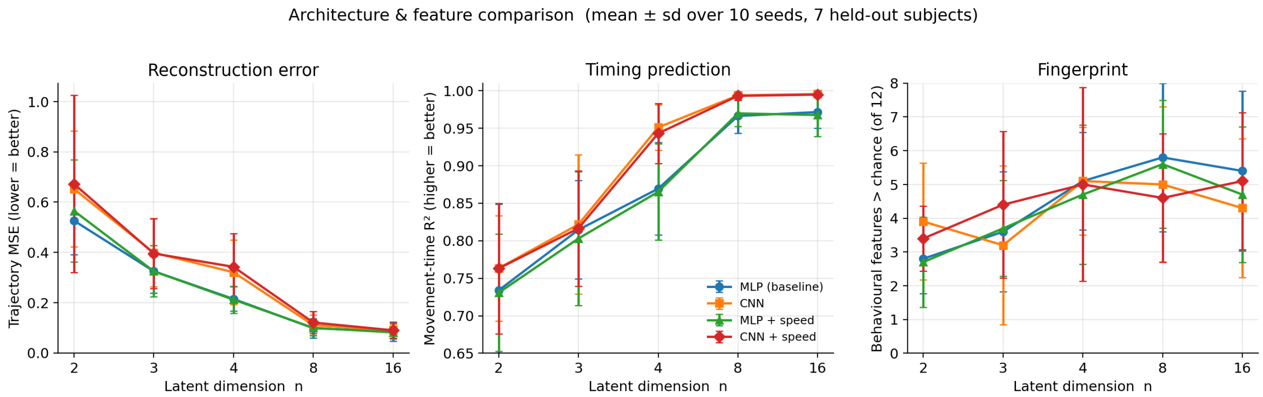

Figure 5. Four variants across latent size (mean ± sd, 10 seeds). The CNN (orange/red) leads on timing; reconstruction and the fingerprint are essentially unchanged.

- **The CNN improves timing prediction** consistently (movement-time R^2 up by 0.03–0.08; it reaches ~0.99 by n=4–8, where the MLP needs n=8+). Respecting frame order helps recover the temporal axis.
- **The CNN does not reconstruct better** — slightly worse at small n, equal at large n — and does not improve the fingerprint.
- **Exact target speed has no measurable effect** on any metric: it duplicates the speed range already carried by the condition.
- **No variant captures the sub-movement strategy**: probing the fingerprint for a subject's sub-movement count gives negative / near-zero R^2 for all variants — the hesitation strategy does not transfer to unseen subjects in the latent.
- **The fingerprint stays data-limited across every architecture and feature** (Figure 5, right).

Conclusion. We adopt the CNN as an optional, principled architecture for the timing / kinematic side; exact-speed conditioning and explicit sub-movement modelling are not worth their added complexity here. Crucially, none of these changes overcomes the fingerprint limitation — independent confirmation that the binding constraint is the 28-subject sample size, not the model. The one lever likely to help the fingerprint would be more subjects.

11. Loss-function experiments

A natural question (raised by Paz) is whether a different training loss could produce better-separated fingerprints. We tested two changes against the standard ELBO, at n=3 and n=8, 10 seeds (Figure 6): a **beta-VAE** (4x the KL weight, for a more structured latent) and a **discriminative** term that explicitly pulls a subject's trials together and pushes different subjects apart. We score them on **held-out fingerprint identification accuracy** — split each unseen subject's trials, build a fingerprint from one half, and check whether the other half lands nearest its own subject (chance = 1/7 ~ 14%).

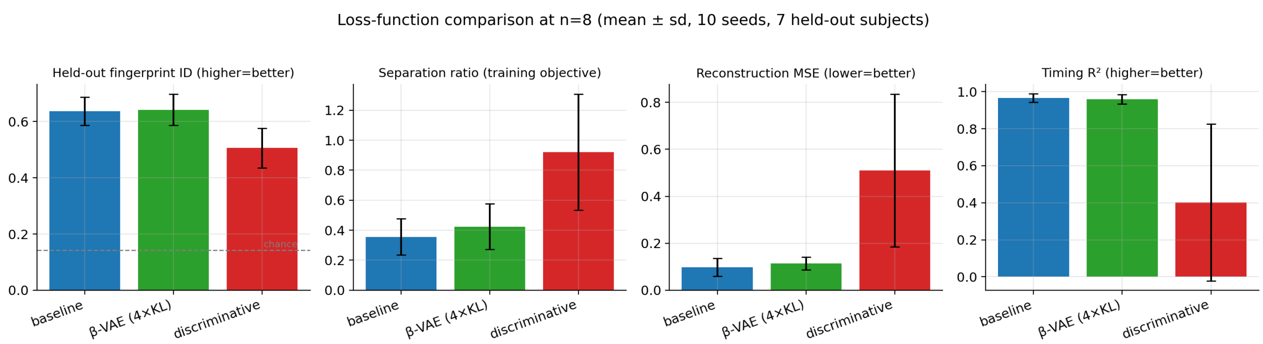

Figure 6. Loss comparison at n=8. The discriminative term inflates the training-set separation ratio but lowers held-out identification and collapses reconstruction and timing.

- **The baseline is already strong** by this metric: it identifies unseen subjects at **~64% at n=8** (vs 14% chance) — the fingerprint carries real, transferable subject signal.
- **beta-VAE: no change** (64.2% vs 63.7%).
- **The discriminative loss backfires:** it raises the training-set separation ratio (0.36 to 0.92) but that does not transfer — held-out identification *drops* to 51%, and reconstruction (MSE 5x) and timing (R^2 0.97 to 0.40) collapse. It memorises the 17 training subjects rather than learning a transferable signature.

Conclusion. Changing the loss does not improve the fingerprint here. The baseline already extracts close to all the subject signal that 28 subjects support, and forcing more separation only overfits the training subjects at the cost of the behavioural information — a third, independent line of evidence that the limit is the data, not the model or its objective.

12. Questions for Prof. Friedman

- **The 'not fixating' flag.** It appears on ~29% of condition-2 (free-eye) trials — more than in condition 1 — and those trials' hand movements look normal. Was there an eye tracker, and is that check meaningful in condition 2? We are currently keeping these trials.
- **Segmentation zero-time.** Do you agree the correct reference is the target's motion onset (the go-signal), and that predicting movement + reaction time addresses the resampling concern?
- **Late-arrival cutoff.** We drop trials arriving >1 s after the target's window as disengagement (there is no clean gap in the data). Is 1 s reasonable?
- **'Too early' trials** (moved before the go-signal) — we exclude these as invalid; please confirm.
- **Sub-movement benchmark.** Could we get access to your sub-movement decomposition pipeline to benchmark generative fidelity against it?